

• Our-I will be scheduled in first week of Feb.

Quotient space :- Example

$$V = \mathbb{R}^2$$

$$W = \text{x axis}$$

$V/W = \{ \text{Set of Equivalence classes of Relation } R \}$

$$v_1, v_2 \in V$$

$$v_1 R v_2 \text{ iff } v_1 - v_2 \in W$$

$$Eq(v_1) = \{ v_1 + w \mid w \in W \} = v_1 + W$$

• Equivalence class of v_1 is same as Equivalence class of v_2 iff $v_1 - v_2 \in W$

$$v \in Eq(v_1) \Rightarrow v R v_1 \Rightarrow v - v_1 \in W$$

$$\Rightarrow v - v_1 = w \text{ for some } w \in W$$

$$\Rightarrow v = v_1 + w$$

$$V = \mathbb{R}^2, W = \text{x axis}$$

$$(x_1, y_1) R (x_2, y_2)$$

$$\text{iff } (x_1, y_1) - (x_2, y_2) \in \text{x axis}$$

$$(x_1 - x_2, y_1 - y_2) \in \text{x axis}$$

$$\text{iff } y_1 = y_2$$

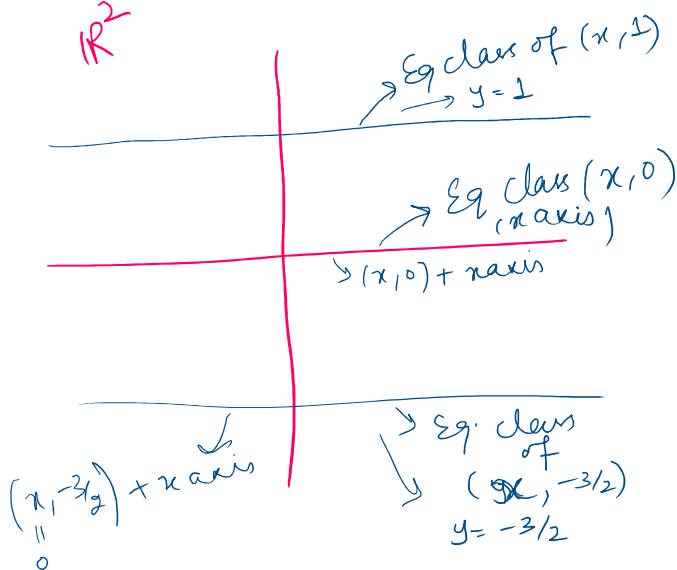
Equivalence class of (x_1, y_1)

$$= \{ (x, y_1) \mid x \in \mathbb{R} \}$$

$$\{ (x, 0) \mid x \in \mathbb{R} \} \rightarrow \text{x axis}$$

$$y_1 = 1$$

$$y_1 = -3/2$$



$$V/W = \{ \text{lines } y = k \mid k \in \mathbb{R} \} = \{ (0, k) + \text{x axis} \mid k \in \mathbb{R} \}$$

$$y = k_1 \quad y = k_2 \quad y = k_1 + k_2$$

$$((0, k_1) + \text{x axis}) + ((0, k_2) + \text{x axis}) = (0, k_1 + k_2) + \text{x axis}$$

First Isomorphism Theorem

Let $T: V \rightarrow W$ be a linear transformation. Then

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Let $T: V \rightarrow W$ be a linear transformation. Then

$$\frac{V}{N(T)} \cong \text{Range}(T)$$

$N(T) :=$ Nullspace of T .

- $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ } Is this a linear transformation?
 $T(x, y) = (0, y)$ } check!

$$\text{Range}(T) = \{T(x, y) \mid (x, y) \in \mathbb{R}^2\} = \{(0, y) \mid y \in \mathbb{R}\} = \text{y axis}$$

$$N(T) = \{(x, y) \in \mathbb{R}^2 \mid T(x, y) = (0, 0)\} = \{(x, y) \in \mathbb{R}^2 \mid (0, y) = (0, 0)\} \\ = \{(x, 0) \mid x \in \mathbb{R}\} = \text{x axis}$$

- By First Isomorphism Thm, we get

$$\frac{\mathbb{R}^2}{\text{x axis}} \cong \text{y axis}$$

$$T: \frac{\mathbb{R}^2}{\text{x axis}} \rightarrow \text{y axis} \quad (\text{bijective L.T.})$$

$$T(\text{line } y=k) = (0, k)$$

$$\parallel \\ (0, k) + \text{x axis}$$

Linear Transformations $\longleftrightarrow$ Matrices

- Matrix associated with a linear transformation

* Let $T: V \rightarrow W$ and $S: V \rightarrow W$ be two linear transformations. Let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be a basis of V . If $T(v_i) = S(v_i) \quad \forall 1 \leq i \leq n$, then

$$T = S$$

T.I.P $T(v) = S(v)$ for all $v \in V$

$$\left. \begin{array}{l} f: A \rightarrow B \quad g: A \rightarrow B \\ f = g \\ \text{iff } f(a) = g(a) \quad \forall a \in A \end{array} \right\}$$

T.L.P $T(v) = S(v)$ for all $v \in V$

$v \in V, v = \alpha_1 v_1 + \dots + \alpha_n v_n$

$$\begin{aligned} T(v) &= T(\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n) \\ &= \alpha_1 T(v_1) + \alpha_2 T(v_2) + \dots + \alpha_n T(v_n) \\ &= \alpha_1 S(v_1) + \alpha_2 S(v_2) + \dots + \alpha_n S(v_n) \\ &= S(\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n) = S(v) \end{aligned}$$

$f = g$
iff $f(a) = g(a) \quad \forall a \in A$

$T, S: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$T(1,0) = S(1,0)$

$T(0,1) = S(0,1)$

T.L.P $T(x,y) = S(x,y) \quad \forall (x,y) \in \mathbb{R}^2$

• $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 $T(x) = (2x, x+y)$

$B = \{ (1,0), (0,1) \}$

$T(1,0) = (2,1) = \check{2}(1,0) + \check{1}(0,1)$

$T(0,1) = (0,1) = \check{0}(1,0) + \check{1}(0,1)$

This map is a linear transformation.

$M_B(T) = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

$T(x,y) = T(x(1,0) + y(0,1))$

$= xT(1,0) + yT(0,1)$

$= x(\check{2}(1,0) + \check{1}(0,1)) + y(\check{0}(1,0) + \check{1}(0,1))$

$= S(x,y)$

• $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$T(x,y,z) = (y, z, y-z)$

$B = \{ \underline{(1,0,0)}, (0,1,0), (0,0,1) \}$

$T(1,0,0) = (0,0,0) = \check{0}(1,0,0) + \check{0}(0,1,0) + \check{0}(0,0,1)$

$T(0,1,0) = (1,0,1) = \check{1}(1,0,0) + \check{0}(0,1,0) + \check{1}(0,0,1)$

$T(0,0,1) = (0,1,-1) = \check{0}(1,0,0) + \check{1}(0,1,0) + \check{(-1)}(0,0,1)$

$M_B(T) = \begin{bmatrix} \check{0} & \check{1} & \check{0} \\ \check{0} & \check{0} & \check{1} \\ \check{0} & \check{1} & \check{-1} \end{bmatrix}$

• $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$T(x,y,z) = (x+y, y+z)$

Check if this map is a linear transformation

$T(x_1, y_1, z_1) + T(x_2, y_2, z_2) = T(x_1+x_2, y_1+y_2, z_1+z_2)$

linear v

$$\begin{aligned}
 \bullet T((x_1, y_1, z_1) + (x_2, y_2, z_2)) &= T(x_1 + x_2, y_1 + y_2, z_1 + z_2) \\
 &= (x_1 + x_2 + y_1 + y_2, y_1 + y_2 + z_1 + z_2) \\
 &= (x_1 + y_1, y_1 + z_1) + (x_2 + y_2, y_2 + z_2) \\
 &= T(x_1, y_1, z_1) + T(x_2, y_2, z_2) \\
 \bullet T(\alpha(x, y, z)) &= T(\alpha x, \alpha y, \alpha z) = (\alpha x + \alpha y, \alpha y + \alpha z) \\
 &= \alpha(x + y, y + z) \\
 &= \alpha T(x, y, z)
 \end{aligned}$$

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x + y, y + z)$

$$B_{\mathbb{R}^3} = \{ (1, 0, 0), (0, 1, 0), (0, 0, 1) \}$$

$$B_{\mathbb{R}^2} = \{ (1, 0), (0, 1) \}$$

$$T(1, 0, 0) = (1, 0) = 1(1, 0) + 0(0, 1)$$

$$T(0, 1, 0) = (1, 1) = 1(1, 0) + 1(0, 1)$$

$$T(0, 0, 1) = (0, 1) = 0(1, 0) + 1(0, 1)$$

$$M_{B_{\mathbb{R}^2}, B_{\mathbb{R}^3}}(T) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3}$$

Question:- $T: V \rightarrow W$ is a L.T.

$$\dim(V) = n, \quad \dim W = m$$

Order of matrix $M(T)$ associated with T
 $m \times n$

• Let $T: V \rightarrow W$ be a linear transformation.

Let $\dim(V) = n$ and $\dim W = m$.

Let $B_V = \{v_1, v_2, \dots, v_n\}$ be a basis of V

• Let $B_W = \{w_1, w_2, \dots, w_m\}$ be a basis of W .

let $\mathcal{B}_V = \{v_1, v_2, \dots, v_n\}$ be a basis of V ,
 and $\mathcal{B}_W = \{w_1, w_2, \dots, w_m\}$ be a basis of W .

Let $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$ be matrix associated with T w.r.t

basis $\mathcal{B}_V$ and $\mathcal{B}_W$, then

$$T(v_1) = a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m = \sum_{i=1}^m a_{i1}w_i$$

$$T(v_2) = a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m = \sum_{i=1}^m a_{i2}w_i$$

$$\vdots$$

$$T(v_n) = a_{1n}w_1 + a_{2n}w_2 + \dots + a_{mn}w_m = \sum_{i=1}^m a_{in}w_i$$

$$1 \leq j \leq n \quad ; \quad T(v_j) = \sum_{i=1}^m a_{ij}w_i$$

• $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$; $T(x, y) = (2x, x+y)$

$\mathcal{B} = \text{Basis} = \{ (1, 0), (1, 1) \}$

$$T(1, 0) = (2, 1) = 1(1, 0) + 1(1, 1)$$

$$T(1, 1) = (2, 2) = 0(1, 0) + 2(1, 1)$$

$$M_{\mathcal{B}}(T) = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\mathcal{B} = \{ (1, 0), (1, 1) \}$$

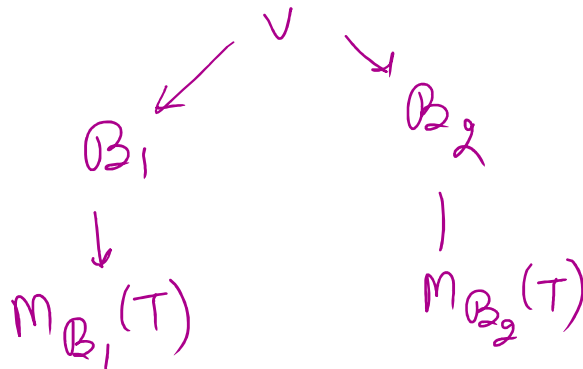
$$M_{\mathcal{B}}(T) = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

• * Matrices of linear T w.r.t different bases are similar matrices.

• Two $n \times n$ matrices A and B are said to be similar $\Rightarrow$ an invertible

- Two $n \times n$ matrices A and B are similar matrices if $\exists$ an invertible $n \times n$ matrix P s.t. $A = P^{-1} B P$.

Let $T: V \rightarrow V$ be linear transformation



$\exists$ an invertible matrix P s.t. $M_{B_1}(T) = P^{-1} M_{B_2}(T) P$.